

REPOSITORY 04

AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

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github.com/johnbarraza/ai-04-acemoglu

1. The paper and the agent's problem

Question

When can accurate, personalized AI improve decisions today yet erode the shared knowledge needed tomorrow?

One mechanism

Human effort jointly produces a private signal and a public externality. AI adds private precision only.

Choice: $e \geq 0$

$$\max_{e \geq 0} \underbrace{f_{00} + G(X)\Delta_G + G(X)G(Y)\Delta_X}_{\text{expected task output}} - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon},$$

$$Y = \sigma^{-2} + \lambda_I e + \tau_A, \quad \Delta_I = 0, \quad \Delta_X > 0.$$

For $X > 0$, the unique interior optimum satisfies

$$\Delta_X G(X) \lambda_I g(Y) = e^{1/\varepsilon}.$$

Source: May 5, 2026 PDF, Sections 3.1–3.5, equations (4)–(6).

2. Main result: one complement, one substitute

Observation 1 (corrected domain). Suppose Gaussian posterior errors, $X > 0$, $Y > 0$, $\lambda_I > 0$, $\tau_A \geq 0$, and Assumption 1 ($\Delta_I = 0$, $\Delta_X > 0$). With $G(z) = 2\Phi(\sqrt{z}) - 1$ and $g = G'$, then

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0$$

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

General knowledge X

Raises the payoff from learning the context: effort and public precision are complements.

Agentic precision τ_A

Enters the same precision index as private learning: AI and effort are substitutes.

Implication. The unique best response $e(X, \tau_A)$ rises in X and falls in τ_A , strictly for $X > 0$.

Source: Observation 1 and Observation 2, pp. 13–15. The paper omits $X > 0$ from the displayed cross-partial domain.

3. What I did: attack the zero-collapse hinge

I restored the production term the paper never relaxes:

$$\Delta_I > 0 \implies U_e = \lambda_I g(Y) \{ \Delta_I + G(X) \Delta_X \} - e^{1/\varepsilon}.$$

At the alleged collapse boundary,

$$U_e(0; 0, \tau_A) = \lambda_I g(\sigma^{-2} + \tau_A) \Delta_I > 0.$$

Hence $e(0, \tau_A) > 0$ and

$$F(0) = [\Sigma^2 + (\lambda_G l e(0, \tau_A))^{-1}]^{-1} > 0.$$

Source: Original extension; baseline Assumption 1 and Sections 5.1–5.3.

Economic consequence

Exact zero knowledge is no longer a fixed point. A low-knowledge trap may survive, but literal collapse does not.

Formal check

EconCSLib target:

AK026KnowledgeCollapse. Lean checks the algebraic/order core; source fidelity remains a separate audit.

4. Where I did not believe the AI

Derivation
observation 1

$$U_{it} = f(\rho, \sigma) + G(x_t) \Delta \theta + G(x_t) G(y_{it}) \Delta v - \frac{\epsilon}{\epsilon+1} e^{\frac{(\epsilon+1)}{\epsilon}}$$

$$\bullet Y = v^{-1} + \lambda_y e + \tau_A$$

derive agents' behavior

$$\frac{\partial U}{\partial e} = \lambda_y \cdot g(y) [\Delta \theta + G(x) \Delta v] - e^{1/\epsilon}$$

Admission on $X=0$, $e=0$ y $G(0)=0$

$$\left. \frac{\partial U}{\partial e} \right|_{\substack{X=0 \\ e=0}} = \lambda_y \cdot g(y) \Delta \theta > 0 \quad \text{si} \quad \sigma > 0, \tau_A > 0, \lambda_y > 0, \Delta \theta > 0$$

But here

$$e^*(0, \tau_A) > 0 \quad \text{y} \quad \text{as } t \rightarrow \infty \text{ el conocimiento público se}$$

$$X_{t+1} = [Z^0 + (\lambda_y I e^*)^{-1}]^{-1} > 0, \quad Z^0 > 0, \lambda_y > 0, \tau_A > 0$$

No se alcanza una frontera de conocimiento hoy pero se aproxima depende de parámetros como $\Delta \theta > 0$

Verdict

The sign calculation is correct on $X > 0$, but it does not identify something unique to an *agent*: any context-specific precision shock added to Y crowds out effort.

The stronger claim—complete destruction of useful knowledge—leans on $\Delta_I = 0$. It is better read as a sharp benchmark for a low-knowledge trap. **Two slips to**

remember:

- ▶ $g(X)$ is not finite at the endpoint $X = 0$;
- ▶ Proposition 5 leaves $\tau_A = \tau_A^C$ unstated.